LATE 1600s: NEWTON

$\infty \leftarrow \bullet \quad \bullet \quad \bullet \quad \bullet \quad \bullet \longrightarrow \infty$

UNIVERSE: ∞ # OF STARS
∞ IN SIZE
STATIC (UNCHANGING)
∞ OLD

1823: OLBERS' PARADOX

IF NEWTON IS CORRECT, EVERY LINE OF SIGHT INTO THE NIGHT SKY MUST END ON A STAR (∞ # OF STARS).

THE LIGHT FROM EVERY STAR IS REACHING US (STATIC, ∞ OLD).

THE NIGHT SKY MUST BE BRIGHT, YET IT APPEARS DARK.

→ AT LEAST ONE OF NEWTON'S IDEAS IS WRONG!

BUT, WHICH ONE(S)??

1915: EINSTEIN'S GENERAL THEORY OF RELATIVITY PREDICTED THE UNIVERSE MUST EITHER EXPAND OR CONTRACT OVER TIME...

→ NOT STATIC!

ADDED A "COSMOLOGICAL CONSTANT" TO REMOVE THIS EFFECT. (REMAIN STATIC).

1929: HUBBLE LAW: $v = H_0 d$

→ THE UNIVERSE IS EXPANDING!

AS OF EARLY 2000s: THERE MAY BE A C.C.!

1940s: "PLAY THE MOVIE BACKWARDS"

STEADY STATE	VS.	BIG BANG
MATTER IS CONSTANTLY BEING CREATED OUT OF THE ENERGY WITHIN SPACE ($E = mc^2$). OLDER MATTER GETS PUSHED OUTWARDS. (∞ OLD)		THE UNIVERSE BEGAN WITH THE EXPANSION OF SPACE IN THE DISTANT PAST. IT HAS BEEN EXPANDING SINCE, CARRYING ALONG THE GALAXIES (NOT ∞ OLD).

1940s: B.B. THEORY: EARLY UNIVERSE WAS VERY HOT, BUT AS IT EXPANDS, IT COOLS...

PREDICTIONS: ① TEMP TODAY SHOULD BE A FEW K. THERE SHOULD BE A COSMIC MICROWAVE BACKGROUND RADIATION (CMBR).

② UNIVERSE SHOULD BE 3/4 H, 1/4 He

1960s: PENZIAS AND WILSON DISCOVERED THE CMBR! TEMP TODAY: 2.73 K ✓

1980s: UNIVERSE IS ABOUT 3/4 H, 1/4 He ✓

3 BIG PIECES OF EVIDENCE SUPPORTING THE B.B.:

1. THE EXPANSION OF THE UNIVERSE
2. COSMIC He ABUNDANCE
3. CMBR *

SOLUTION TO OLBERS' PARADOX:

NEWTON'S UNIVERSE: STATIC — NOPE!
∞ OLD — NOPE!
∞ IN SIZE — NOPE!
∞ # OF STARS — NOPE!

GOAL: USING THE LAWS OF PHYSICS, WE WANT TO DESCRIBE THE NATURE OF THE UNIVERSE ALL THE WAY BACK TO TIME Ø (INSTANT OF THE BIG BANG).

TEST THEORIES? USE PARTICLE ACCELERATORS)

AS WE GO BACK, THE 4 FORCES OF NATURE (GRAVITY, ELECTROMAGNETISM, WEAK & STRONG FORCE) BEGIN TO UNIFY (ACT LIKE A SINGLE FORCE).

ASSUMPTIONS ABOUT THE UNIVERSE:

1. HOMOGENEOUS (EVERY PLACE LOOKS THE SAME)
2. ISOTROPIC (SAME IN ALL DIRECTIONS)
3. OBEYS "UNIVERSALITY" (EVERYTHING IS MADE OF THE SAME STUFF AND OBEYS THE SAME LAWS OF PHYSICS).

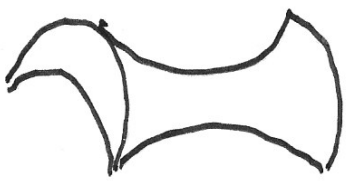
SO...

4. COSMOLOGICAL PRINCIPLE: NO SPECIAL PLACE EXISTS IN THE UNIVERSE. IF THIS IS TRUE,

CAN THERE BE: A UNIQUE CENTER? EDGE? NOPE!

SHAPE OF THE UNIVERSE?

FLAT

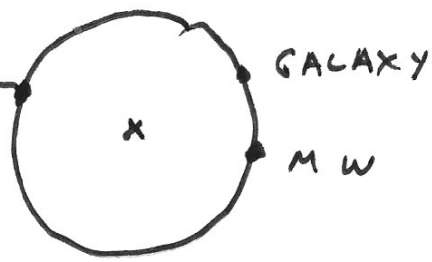


NEGATIVE

✓

POSITIVE

OUR UNIVERSE MAY BE THE SURFACE <u>ONLY</u> OF A 4 PHYSICAL DIMENSIONAL BALL (HYPERSPHERE)



EXPANDING UNIVERSE? BALL GETS BIGGER.

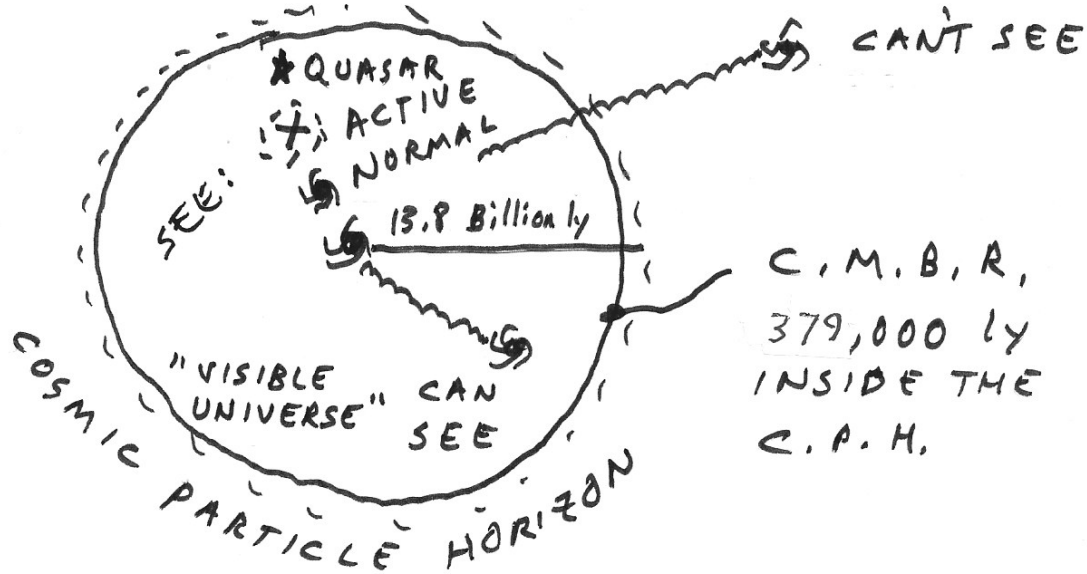
CENTER? OF UNIVERSE
- 3D: EVERYWHERE!
- 4D: NOWHERE!

HUBBLE LAW: $v = H_0 d$ $\quad$ v = RECESSIONAL VELOCITY, d = DISTANCE

$$\frac{d}{v} = \frac{vt}{v} \rightarrow t = \frac{d}{v} \qquad \frac{v}{H_0 v} = \frac{H_0 d}{H_0 v} \rightarrow \frac{1}{H_0} = \frac{d}{v}$$

IF $H_0 \approx 71 \frac{km}{s \cdot MP}$

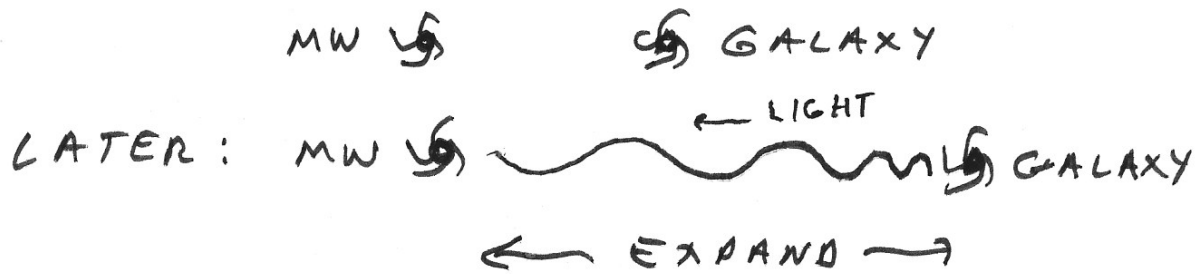
$\frac{1}{H_0}$ = AGE OF THE UNIVERSE! AGE: 13.8 BILLION YEARS



WE ARE AT THE CENTER OF OUR VISIBLE UNIVERSE, BUT NOT AT THE CENTER OF THE ENTIRE UNIVERSE.

HUBBLE LAW: DOPPLER SHIFT (RECESSIONAL VELOCITY)

TODAY: COSMOLOGICAL REDSHIFT!



TODAY, USING THE LAWS OF PHYSICS, WE CAN DESCRIBE THE NATURE OF THE UNIVERSE ALL THE WAY BACK TO TIME .... 10^{-43} s

$10^{-43} = 0.0000000000000000000000000000000000000001$

① DANG CLOSE TO 0!

② AIN'T 0!